HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Mathematics

F18CE

Multivariable Calculus and Real Analysis B

Semester 2 – 2020/21

Attempt all questions

Appropriate working must be shown to obtain full credit.

This exam contributes 70% of the overall mark for this course. There are 70 marks available (that is, 1 mark is worth 1%).

Attempt all questions **Show any calculations you make**
(no marks will be given if there is no, or incorrect, working, even if the answer is correct).

If a question says to 'use the definition' to prove some result, your answer must rely solely on the definition to prove the result, and not on other results from the course.
If the question does not say this then you may use any result from the course in your answer.

1 Let (a_n) be a sequence of real numbers.

Using the definition of the limit of a sequence, prove the following results.

(a) $\lim_{n\to\infty} \dfrac{3n^2 + 2n\sin n}{n^2 + 4n + 5} = 3\,.$

(b) If $\lim_{n\to\infty} a_n = L > 0$ then there exists an integer N such that

$$|a_n| < 1 + L, \quad \text{for all } n \geq N.$$

(c) If $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$ then $\lim_{n\to\infty} (a_n + b_n) = L + M.$ **[9 marks]**

2 Give examples of sequences, one in each case, of the following.

(a) (a_n) is unbounded, but has a convergent subsequence.

(b) (a_n) is not convergent but (a_n^2) is convergent.

(c) (a_n) is not convergent, but $(a_n^2 - 3a_n + 2)$ is convergent. **[6 marks]**

3 Let A be a non-empty, bounded set of positive real numbers. Let B be the set consisting of numbers of the form

$$b = a^3 + 5, \quad \text{for } a \in A.$$

Show that B is bounded above, and that

$$\sup B = (\sup A)^3 + 5.$$ **[6 marks]**

4 (a) Give an example of a bounded sequence having subsequences converging to at least three different limits.

(b) Suppose that the sequence (a_n) is defined by

$$a_n = \frac{3n^2 + n\sin(n^2) + 6}{2n^2 + n\cos(n^2) + 1}, \quad n \geqslant 1.$$

Show that (a_n) has a convergent subsequence. **[5 marks]**

5 Suppose that $f : A \to \mathbb{R}$ is a function defined on an open interval A, and $a \in A$. Using the definition of the limit of a function, prove that

$$\lim_{x\to 2} (x^2 + 3x) = 10.$$ **[6 marks]**

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6 Suppose that the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are continuous at a.

Show that the function $f \times g : \mathbb{R} \to \mathbb{R}$ is continuous at a
(you may use standard theorems about limits here). **[3 marks]**

7 Show that the equation

$$6e^x = 3e^{-2x} + 2$$

has a unique solution on $\mathbb{R}$. **[6 marks]**

8 Suppose that the functions $f,\ g : \mathbb{R} \to \mathbb{R}$ are defined by

$$f(x) = \begin{cases} x^2, & x < 0, \\ 2x^2 + 1, & x \geq 0, \end{cases}$$

$$g(x) = x^2 f(x), \quad x \in \mathbb{R}.$$

Use the definition of differentiability to prove that:

(i) the function f is not differentiable at 0;

(ii) the function g is differentiable at 0 **[6 marks]**

(you must use the definition of differentiability, but you may use standard theorems about limits here).

9 Use the First Mean Value Theorem to prove that

$$\tan x > x, \quad \text{for } x \in (0, \pi/2). \qquad \textbf{[3 marks]}$$

10 (a) Consider the series $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^\alpha}$. For which values of $\alpha > 0$ does it converge and for which does it diverge?

(b) Determine the convergence of the series $\displaystyle\sum_{r=1}^{\infty} a_r$ for the following cases:

$$(i) \quad a_r = \frac{3r^2 - r + 5}{2r^4 + r^2 + 10}; \qquad (ii) \quad a_r = \frac{r^3 + 3r - 2}{r^4 + 2r^3 + 4}. \qquad \textbf{[9 marks]}$$

11 Find the radius of convergence R of the power series

$$\sum_{r=0}^{\infty} \frac{x^r}{3^r \sqrt{r^2 + 3}}. \qquad \textbf{[3 marks]}$$

12 Let $g : [1, 4] \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} 2, & 1 \leqslant x < 3, \\ 7, & x = 3, \\ -1, & 3 < x \leqslant 4. \end{cases}$$

For any integer $n \geqslant 2$, let P_n be the partition given by $P_n = \left\{1, 3 - \frac{1}{n}, 3 + \frac{1}{n}, 4\right\}$.

Find the upper and lower sums $U(P_n, g)$ and $L(P_n, g)$.

Hence show that g is integrable on $[1, 4]$ and find $\displaystyle\int_1^4 g(x)\, dx$. **[8 marks]**

END OF EXAM

1 (a) Let $\epsilon > 0$.

$$\left|\frac{3n^2 + 2n\sin n}{n^2 + 4n + 5} - 3\right| = \left|\frac{2n\sin n - 3(4n+5)}{n^2 + 4n + 5}\right| \leqslant \frac{29n}{n^2} < \epsilon \quad \text{if} \quad n > \frac{29}{\epsilon}.$$

So if we choose N such that $N > 29/\epsilon$, then

$$|a_n - L| < \epsilon \quad \text{for all } n \geq N.$$ (3)

(b) Let $\epsilon = 1$. Choose N such that

$$n \geqslant N \implies |a_n - L| < 1 \implies |a_n| = |a_n - L + L| \leqslant |a_n - L| + L < 1 + L.$$ (3)

(c) Let $\epsilon > 0$.
Choose N_a such that $n \geqslant N_a \implies |a_n - L| < \frac{1}{2}\epsilon$.
Choose N_b such that $n \geqslant N_b \implies |b_n - L| < \frac{1}{2}\epsilon$.
Then, letting $N = \max\{N_a, N_b\}$, gives

$$n \geqslant N \implies |a_n + b_n - L - M| \leqslant |a_n - L| + |b_n - M| < \frac{1}{2}\epsilon + \frac{1}{2}\epsilon = \epsilon.$$ (3)

2 (a) $a_n = 1, 0, 2, 0, 3, 0 \ldots$; (2)

(b) $a_n = (-1)^n$; (2)

(c) $a_n = 1, 2, 1, 2, \ldots$. (2)

3 ▸ By the definition of $\sup A$, for all $a \in A$:

$$0 \leqslant a \leqslant \sup A \implies 0 \leqslant a^3 + 5 \leqslant (\sup A)^3 + 5.$$

Hence, B is bounded and $(\sup A)^3 + 5$ is an upper bound for B. (3)

▸ We can choose a sequence (a_n) in A such that $a_n \to \sup A$, and hence $a_n^3 + 5 \to \sup A^3 + 5$. Since $a_n^3 + 5 \in B$, for all $n \geqslant 1$, combining this with the upper bound on B from the previous part shows that $\sup B = \sup A^3 + 5$. (3)

4 (a) $a_n = 1, 0, -1, 0, 1, 0, -1, \ldots$; (2)

(b)

$$|a_n| \leqslant \frac{|3n^2 + n\sin(n^2) + 6|}{|2n^2 + n\cos(n^2) + 1|} \leqslant \frac{10n^2}{n^2} \leqslant 10,$$

so (a_n) is bounded, and so by BWT has a convergent subsequence. (3)

5 Let $\epsilon > 0$. If $|x - 2| < 1$ then

$$|x - 2| < 1 \implies -1 < x - 2 < 1 \implies 1 < x < 3 \implies |x| < 3,$$

$$|x^2 + 3x - 10| = |(x+5)(x-2)| \leqslant (|x| + 5)|x - 2| < 8|x - 2|.$$ (3)

Now let

$$\delta = \min\left\{1, \frac{\epsilon}{8}\right\}.$$

Then:

$$|x^2 + 3x - 10| < 8|x - 2| < \epsilon \quad \text{if } 0 < |x - 2| < \delta,$$

so that, by definition,

$$\lim_{x \to 2}(x^2 + 3x) = 10.$$ (3)

6 Since f and g are continuous, $\lim_{x\to a} f(x) = f(a)$, $\lim_{x\to a} g(x) = g(a)$, so by a standard theorem on limits

$$\lim_{x\to a}(f(x)\times g(x)) = f(a)\times g(a),$$

which shows that $f \times g$ is continuous. (3)

7 Let $f(x) = 6e^x - 3e^{-2x} - 2$, $x \in \mathbb{R}$.

Then f is continuous and differentiable on $\mathbb{R}$.

- Now,
$$f(1) = 6e - 3e^{-2} - 2 > 1, \quad f(-1) = 6e^{-1} - 3e^2 - 2 < -2,$$
so by the IVT there exists $c \in [-1, 1]$ such that $f(c) = 0$, that is, $6e^c = 3e^{-2c} + 2$. (3)
- Also,
$$f'(x) = 6e^x + 6e^{-2x} > 0 \quad \text{on } \mathbb{R},$$
so there cannot be 2 solutions by Rolle's theorem (3)
[not needed for marks, but argument is: if there were two solutions, say $a < b$, then Rolle's theorem would imply that $f'(c) = 0$, for some $a < c < b$, which is a contradiction].

8 Put $a = 0$ and $h = x - a = x$.

(i) For f, the left hand limit at $a = 0$ is:

$$h < 0 \implies \frac{f(h) - f(0)}{h} = \frac{h^2 - 1}{h} \to -\infty,$$

so the limit does not exist and so f is not differentiable at 0. (3)

(ii) For g, the left and right hand limits are:

$$h < 0 \implies \frac{g(h) - g(0)}{h} = \frac{h^4}{h} \to 0,$$
$$h > 0 \implies \frac{g(h) - g(0)}{h} = \frac{h^2(2h^2 + 1)}{h} \to 0.$$

The left and right hand limits are equal, so the limit exists and g is differentiable at 0. (3)

9 Define $f(x) = \tan x - x$ for $x \in [0, \pi/2)$. Then f is continuous on $[0, \pi/2)$, and differentiable on the open interval $(0, \pi/2)$, and

$$f'(x) = \frac{1}{\cos^2 x} - 1 > 0, \quad \text{for } 0 < x < \pi/2.$$

By the Mean Value Theorem, on the interval $[0, x]$, with $0 < x < \pi/2$, there exists $c \in (0, x)$ such that

$$f(x) = f(0) + f'(c)x > 0 \implies \tan x - x > 0 \quad \text{for } x \in (0, \pi/2).$$ (3)

10 (a) The given series converges if $\alpha > 1$ and diverges if $\alpha \leqslant 1$. (3)

(b) (i) For $r \geqslant 1$, $a_r \geqslant 0$, and $a_r = \dfrac{3r^2 - r + 5}{2r^4 + r^2 + 10} \leqslant \dfrac{8r^2}{2r^4} = 4\,\dfrac{1}{r^2}$

(making the top line bigger and the bottom line smaller)
and the series $1/r^2$ converges, so the original series converges. (3)

(ii) For $r \geqslant 1$, $a_r = \dfrac{r^3 + 3r - 2}{r^4 + 2r^3 + 4} \geqslant \dfrac{r^3}{7r^4} = \dfrac{1}{7}\,\dfrac{1}{r}$

(making the top line smaller and the bottom line bigger)
and the series $\sum 1/r$ diverges, so the original series diverges. (3)

11 By the Ratio test,

$$\frac{3^r\sqrt{r^2+3}}{3^{r+1}\sqrt{(r+1)^2+3}}\,|x| = \frac{1}{3}\,|x|\left(\frac{r^2+3}{(r+1)^2+3)}\right)^{1/2}|x| \to \frac{1}{3}\,|x| \qquad \text{for any } x,$$

so the power series converges for any $|x|<3$, and diverges for $|x|>3$ so the radius of convergence is $R=3$. **3**

12 For the given function g and partitions P_n:

$$\sup\left\{g(x) : x \in \left[1, 3-\frac{1}{n}\right]\right\} = 2\,,$$

$$\sup\left\{g(x) : x \in \left[3-\frac{1}{n}, 3+\frac{1}{n}\right]\right\} = 7\,,$$

$$\sup\left\{g(x) : x \in \left[3+\frac{1}{n}, 4\right]\right\} = -1$$

so that

$$U(P_n,g) = 2\left(\left(3-\frac{1}{n}\right)-1\right)+7\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right)-1\left(4-\left(3+\frac{1}{n}\right)\right) = 3+\frac{13}{n}\cdot$$

Similarly,

$$L(P_n,g) = 2\left(\left(3-\frac{1}{n}\right)-1\right)-1\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right)-1\left(4-\left(3+\frac{1}{n}\right)\right) = 3-\frac{3}{n}\cdot$$

Thus

$$\lim_{n\to\infty}\left(U(P_n,g)-L(P_n,g)\right) = \lim_{n\to\infty}\frac{16}{n} = 0.$$

Hence g is integrable on $[1,4]$ and

$$\int_1^4 g(x)\,dx = \lim_{n\to\infty} U(P_n,g) = 3.$$ **8**